

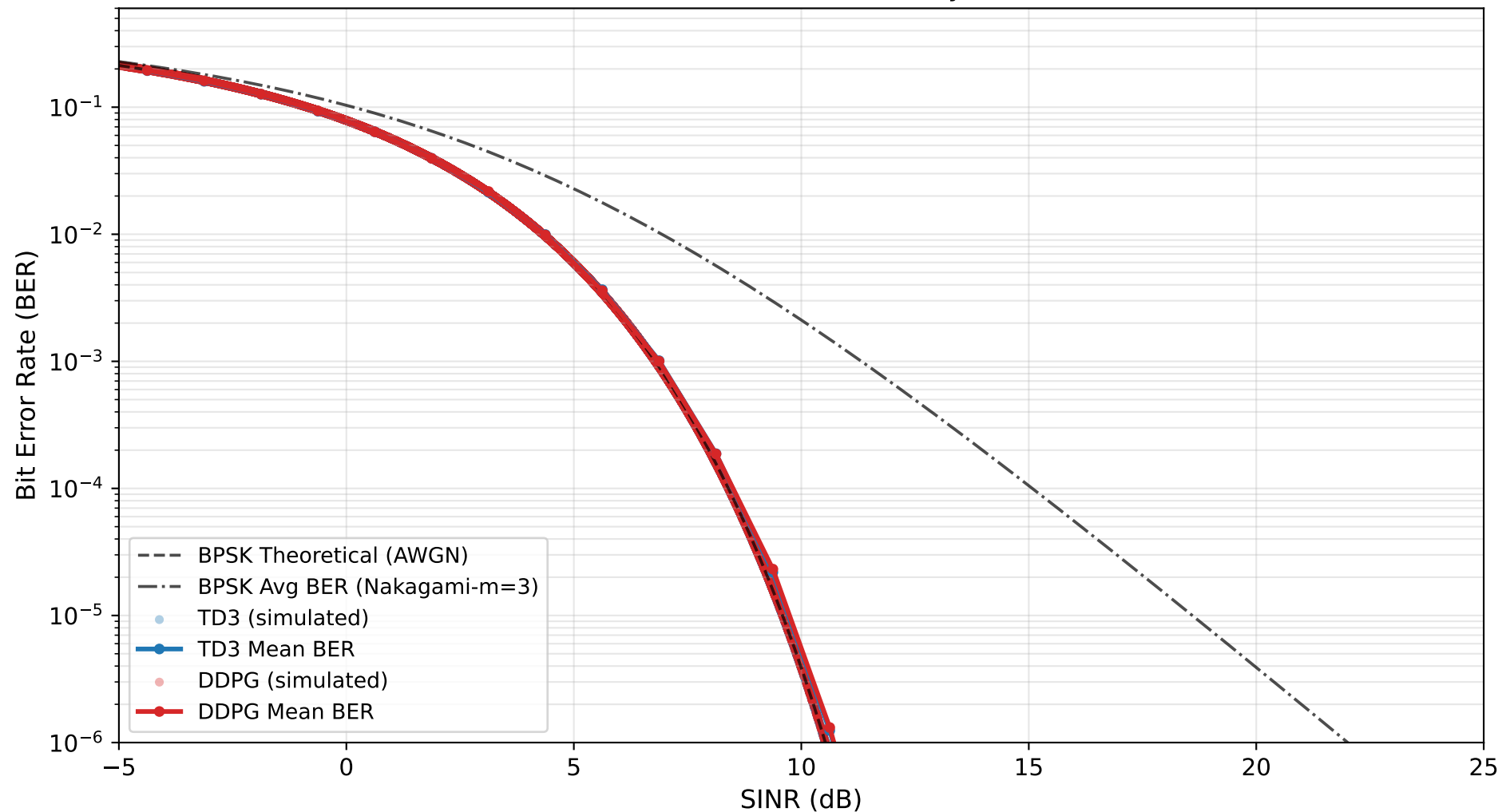
CRN Power Control: TD3 vs DDPG

Nakagami-m Fading Channel Performance Report

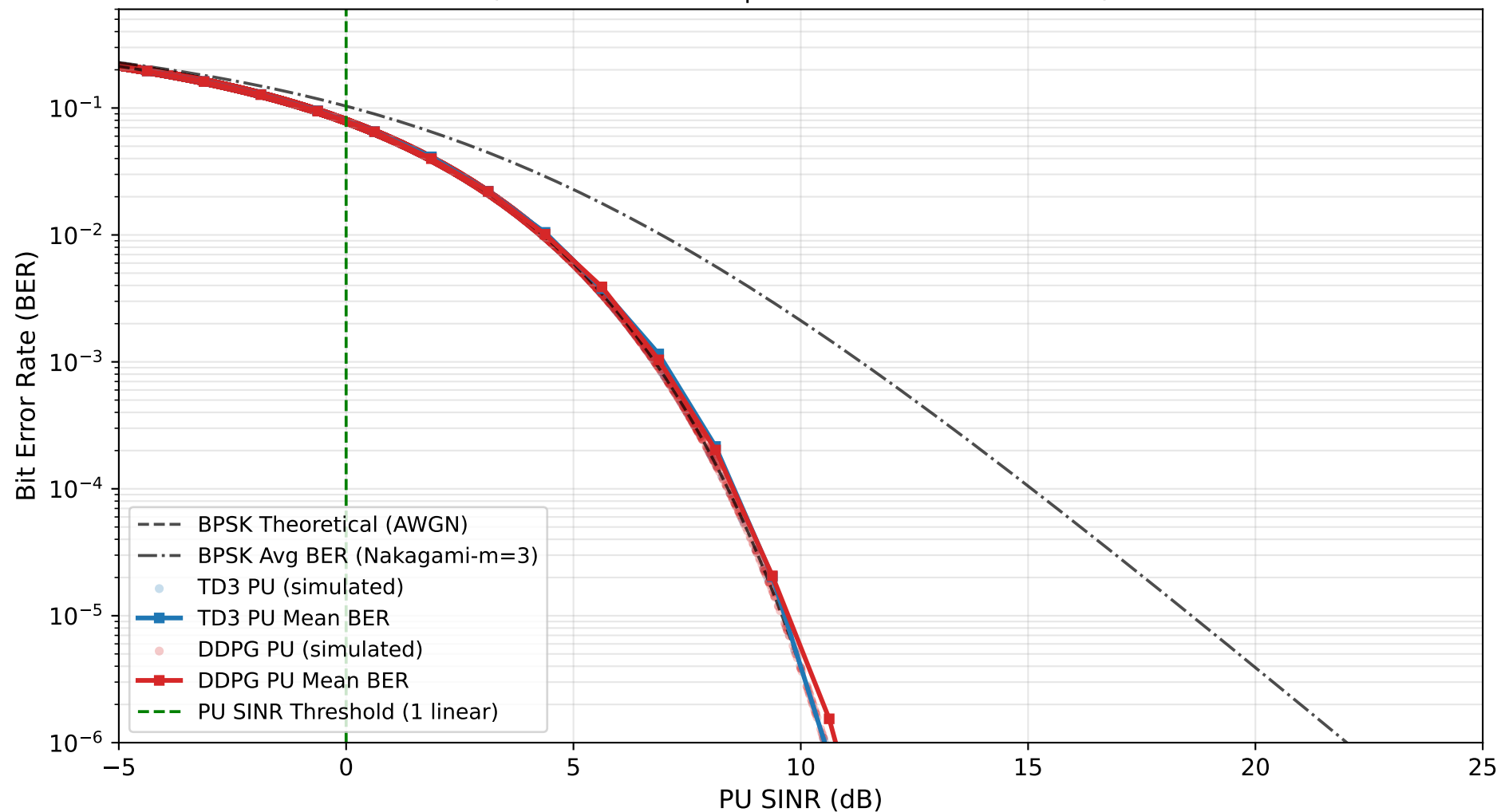
Nakagami-m = 3.0 | Episodes = 3500 | Steps/Episode = 200 | SINR Threshold = 1.0 dB

Metric	TD3	DDPG	Winner
Avg Reward (last 100 ep)	241.2366	209.1088	TD3
SU Throughput (bits/s/Hz)	2.0669	1.6934	TD3
PU Throughput (bits/s/Hz)	0.5311	0.7744	DDPG
Outage Probability	0.1132	0.2508	TD3
Average BER	0.028006	0.053370	TD3
Training Time (s)	74679.6	43462.4	-

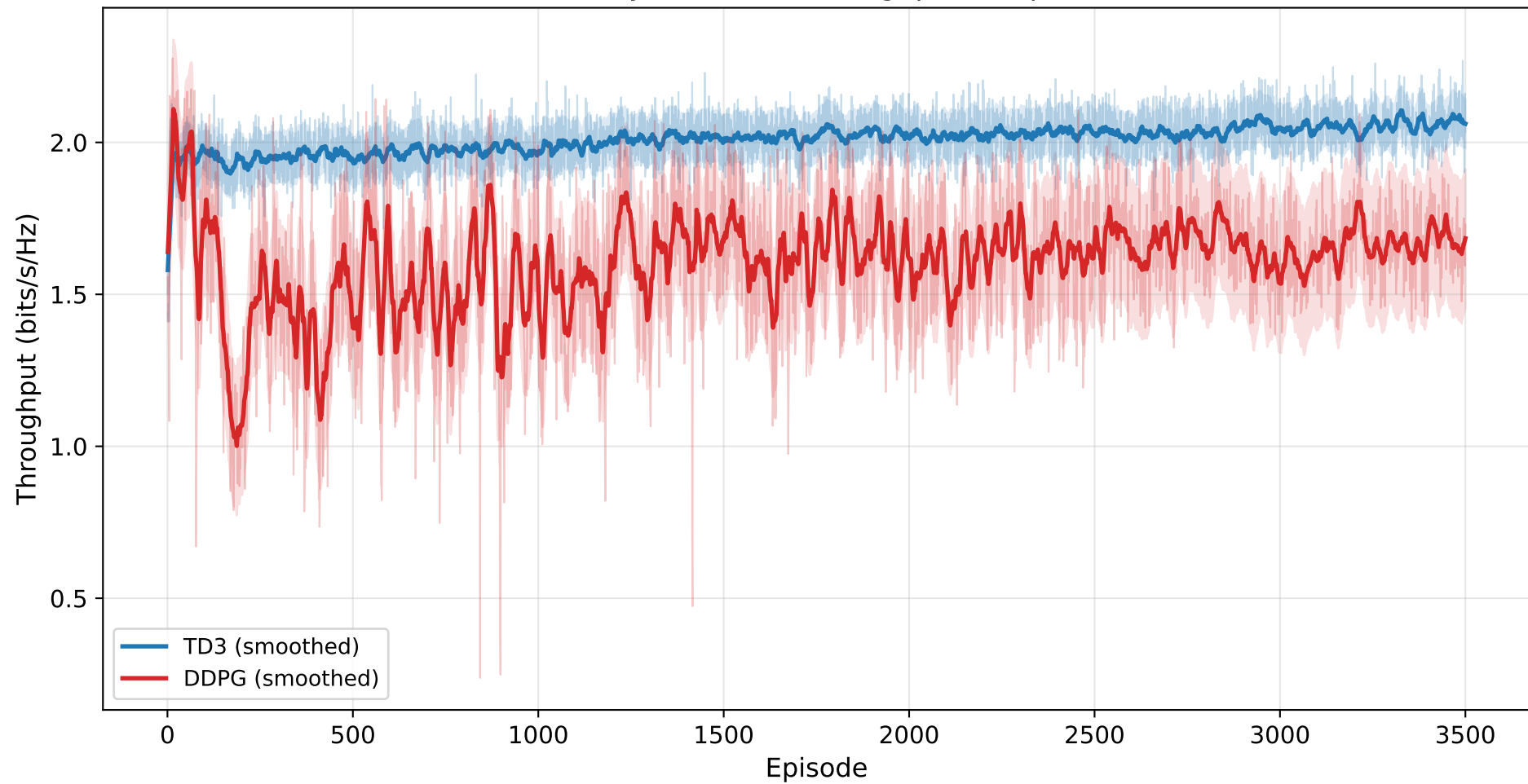
SINR vs BER - Nakagami-m=3 Fading Channel
(BPSK Modulation, Secondary User)



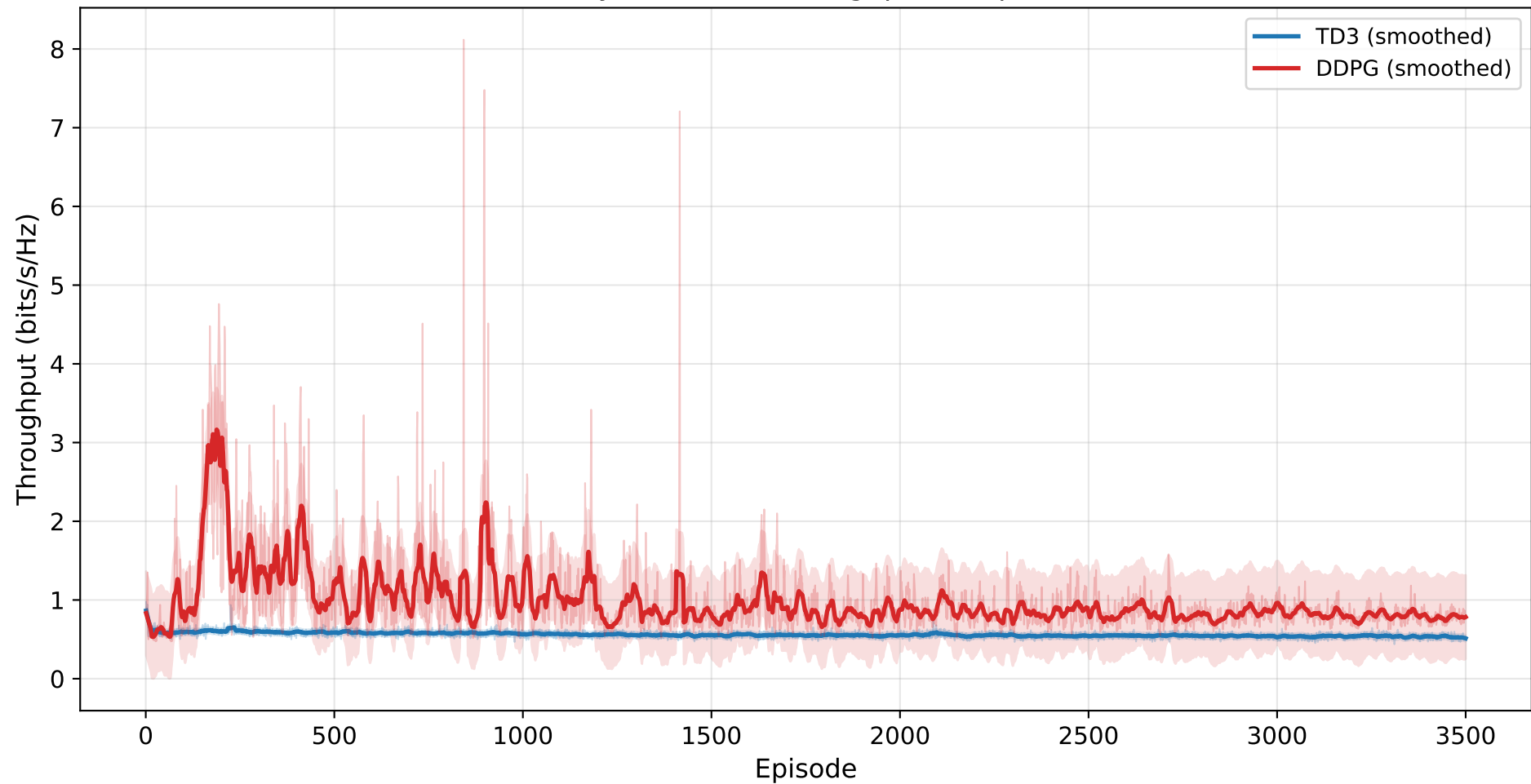
Primary User SINR vs BER - Nakagami-m=3 Fading Channel (BPSK Modulation | Effect of SU Interference)



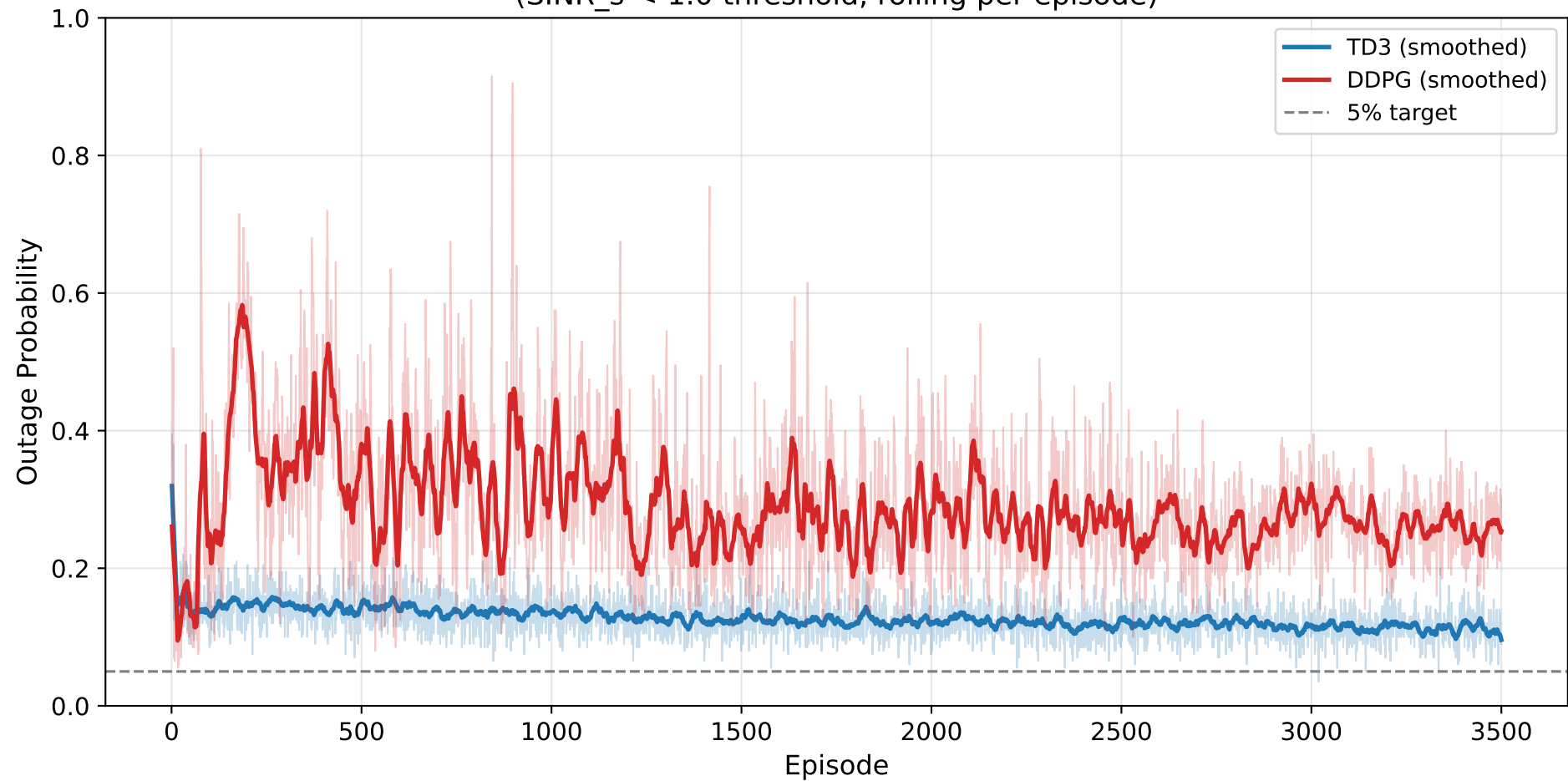
Secondary User (SU) Throughput vs Episodes



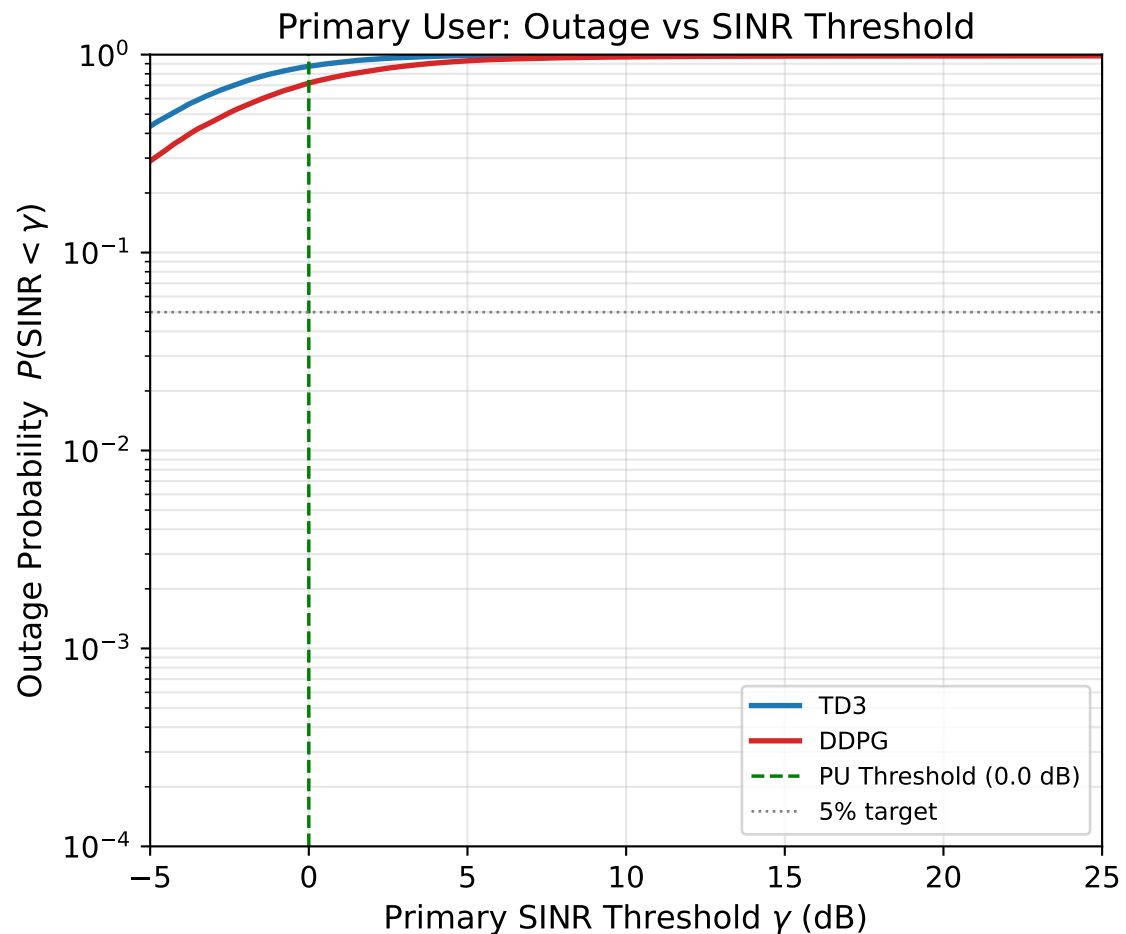
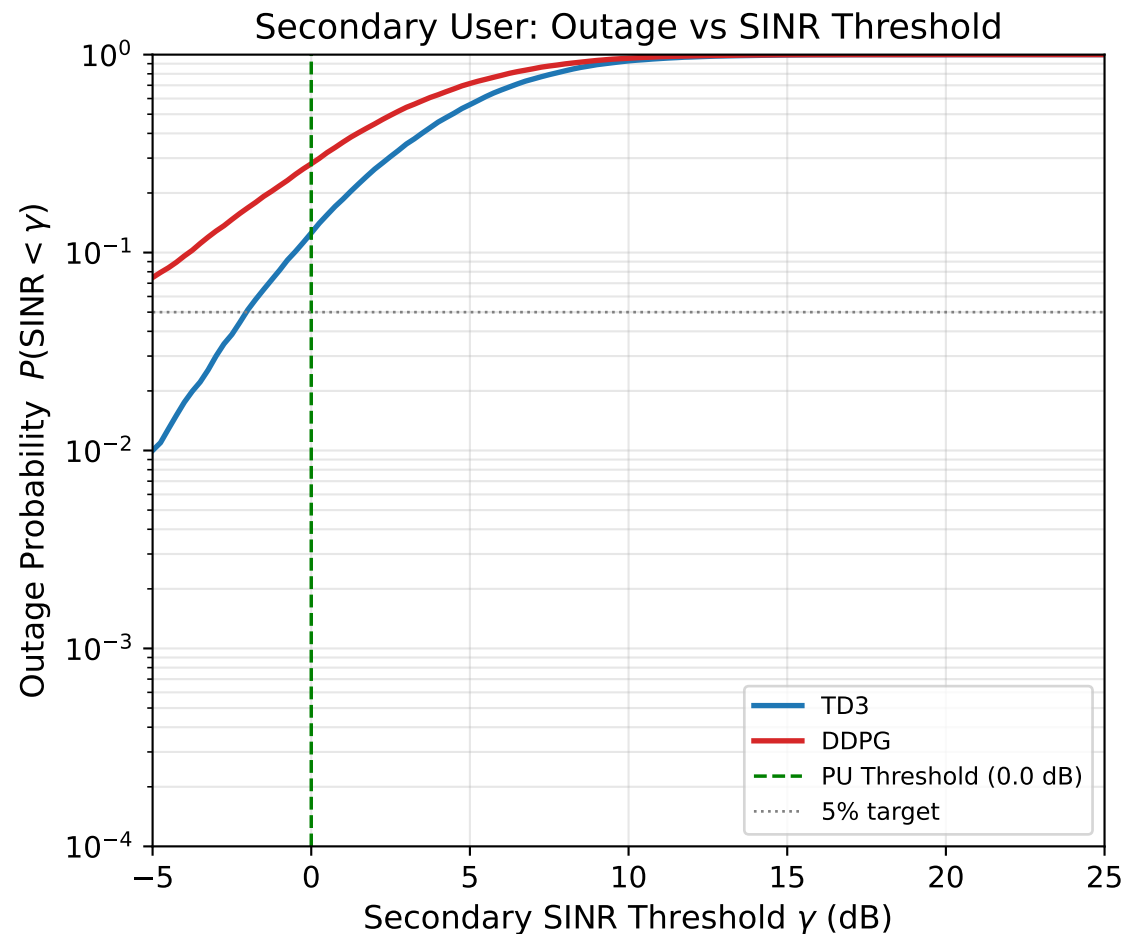
Primary User (PU) Throughput vs Episodes



Outage Probability vs Episodes
(SINR_s < 1.0 threshold, rolling per episode)

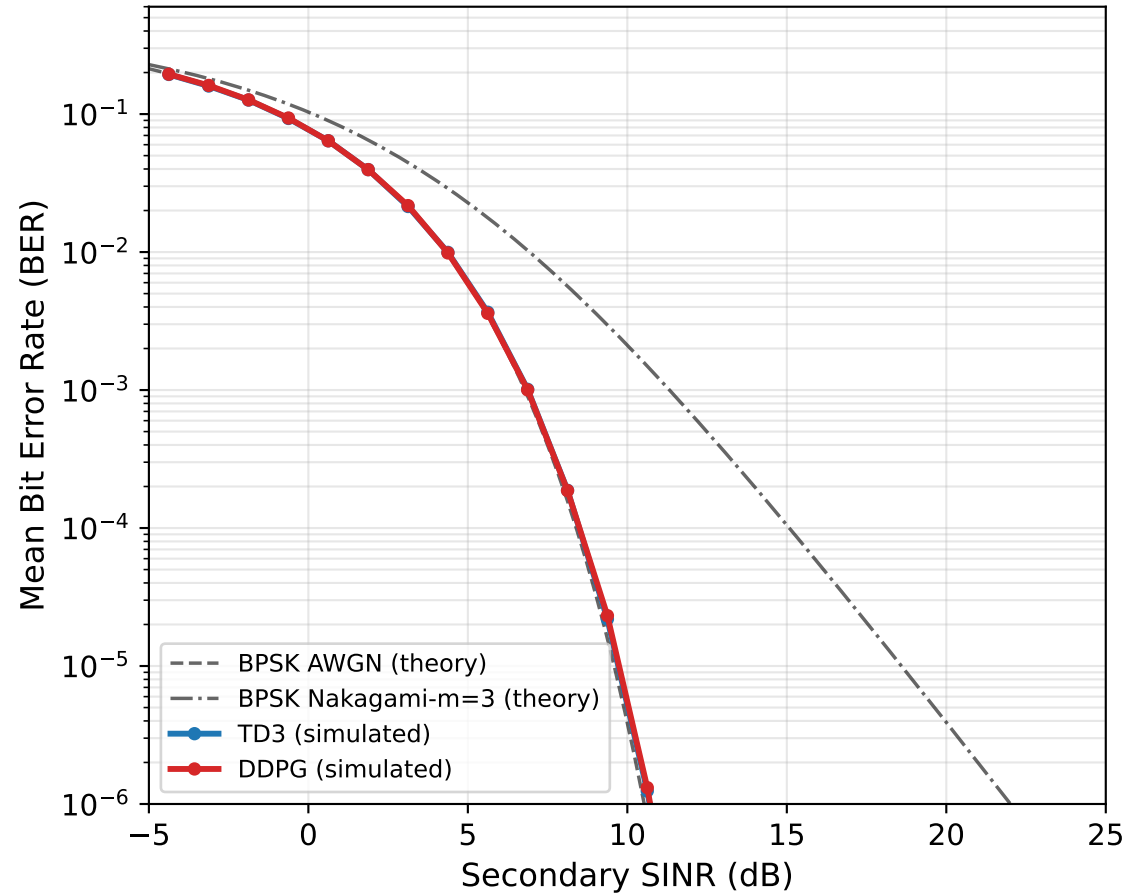


Outage Probability vs SINR Threshold (Empirical CDF) — Imperfect CSI

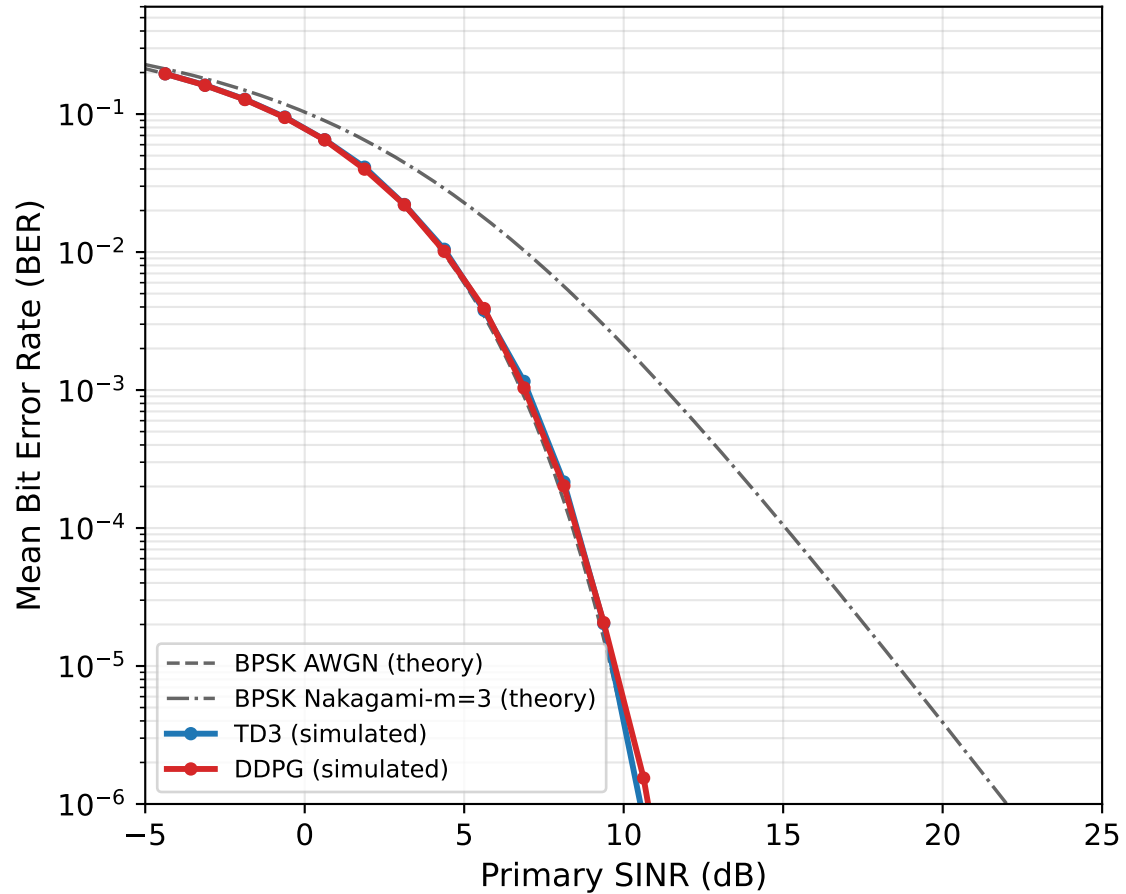


BER vs SINR — Algorithm Comparison under Imperfect CSI

Secondary User: BER vs SINR

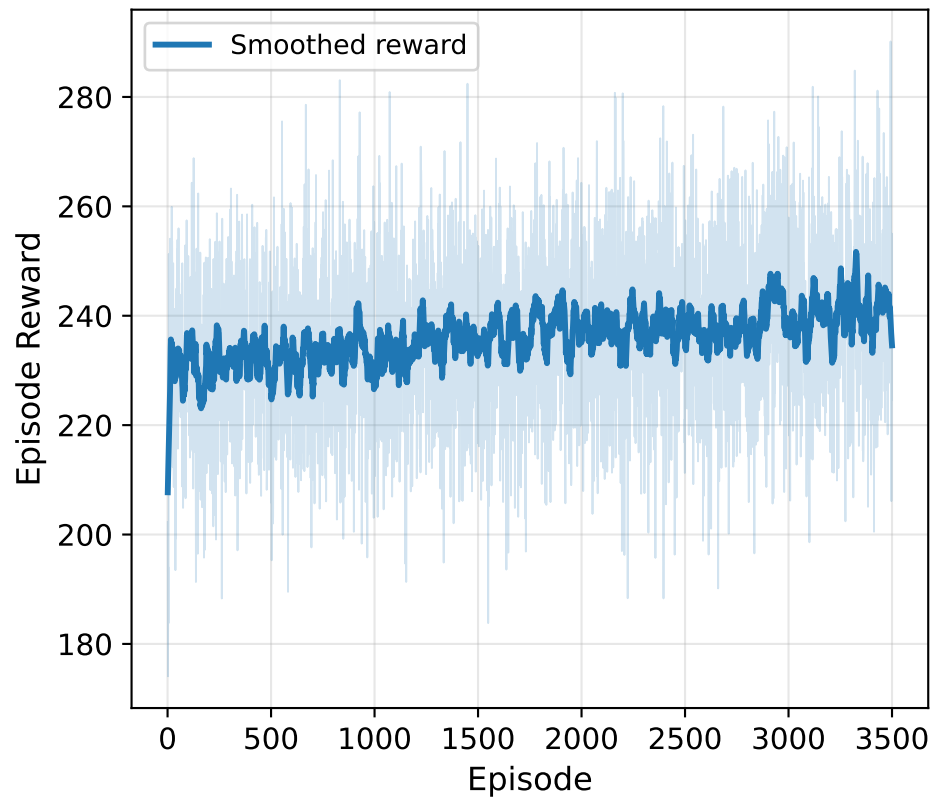


Primary User: BER vs SINR

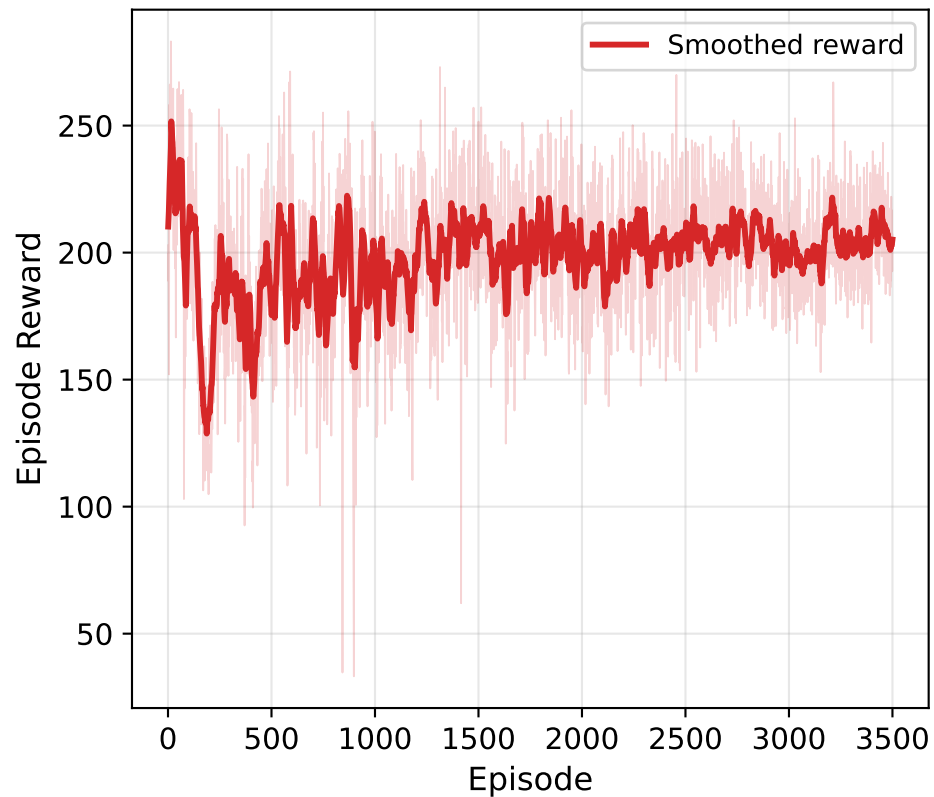


Episode Reward Curves - TD3 vs DDPG

TD3 Reward Curve



DDPG Reward Curve



TD3 vs DDPG - Reward Comparison

